

Bluestock Mutual Fund Analytics Platform

FINAL PROJECT REPORT

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Bluestock Fintech
Mutual Fund Data Analytics

Python • SQL • SQLite • Power BI

40 Schemes • 10 Fund Houses • ETL • Star Schema • Advanced Analytics

Final submission report

1. Executive Summary

The Bluestock Mutual Fund Analytics Platform is an end-to-end data analytics project developed during the Bluestock Fintech capstone. The platform converts multiple mutual fund datasets into a structured analytical environment and provides a business-facing dashboard for exploring industry trends, scheme performance, investor activity and advanced risk indicators.

The solution covers 40 real mutual fund schemes belonging to 10 fund houses. The analytical environment contains approximately 46,000 NAV observations, 32,778 investor transaction records, 90 fund-house AUM observations, 40 scheme-performance records, 322 portfolio-holding records and 8,050 benchmark observations. Additional monthly datasets capture SIP inflows, category inflows and folio information.

The project follows a reproducible workflow: data ingestion → validation → cleaning → transformation → relational storage → exploratory analysis → performance and risk analysis → advanced analytics → Power BI visualization. Each stage has dedicated scripts, notebooks or database objects, making it possible to trace outputs back to their underlying datasets.

The performance layer calculates CAGR-till-date, Sharpe ratio, Sortino ratio, alpha, beta and maximum drawdown. Advanced analytics extends this with historical VaR/CVaR, rolling 90-day Sharpe, investor cohorts, SIP continuity, a rule-based recommendation prototype and sector concentration using HHI.

The Power BI deliverable provides four primary pages—Industry Overview, Fund Performance, Investor Analytics and SIP & Market Trends—and a Fund Details drill-through page. Slicers, navigation and DAX measures allow users to move from high-level indicators to scheme-level analysis.

A key feature of the report is transparency about data provenance. Fund metadata, AMFI codes, historical NAV and benchmark information are based on public-source data where applicable. Investor transactions and portfolio holdings are synthetic or constructed for the capstone, and constructed datasets are not represented as official investor or portfolio disclosures.

2. Project Background

2.1 Business Context

Mutual fund analysis requires information from several dimensions at once. A scheme has descriptive characteristics such as category, plan, benchmark and expense ratio, while its historical NAV provides a time series for performance analysis. Industry-level AUM and SIP flows provide a broader market view, while investor transactions provide a behavioral view.

When these datasets are maintained separately, analysts may spend significant time cleaning dates, aligning identifiers, checking missing values and building repeated calculations. A centralized analytics platform reduces this fragmentation by standardizing the data pipeline and defining consistent analytical measures.

2.2 Problem Statement

The project problem is to design and implement a reliable mutual fund analytics platform that can ingest heterogeneous datasets, validate and clean them, store them in a relational structure, generate meaningful performance and investor metrics, and expose the results through an interactive dashboard.

2.3 Analytical Questions

- How has industry AUM changed over the available time period?
- How have SIP inflows, SIP accounts and folios changed?
- Which schemes show stronger historical returns and risk-adjusted metrics?
- How large are historical drawdowns across schemes?
- How do investor age, geography, gender and transaction types differ?
- How continuous are SIP contributions under the project's continuity rule?
- How concentrated are constructed portfolio holdings across sectors?
- How can a transparent rule-based scoring prototype support fund comparison?

2.4 Project Scope

The scope covers ETL, SQL database design, EDA, performance analytics, advanced analytics and Power BI reporting. It does not attempt to provide regulated financial advice or predict future market performance.

3. Objectives and Requirements

Objective	Detailed Implementation	Evidence / Output
O1	Automated ingestion, cleaning and validation	Python scripts + processed CSVs
O2	Relational star-schema database	SQLite DB + schema.sql
O3	Business-focused EDA	03_eda_analysis.ipynb + charts
O4	Performance and risk analytics	04_performance.ipynb + metric CSVs
O5	Interactive dashboard	Power BI PBIX + PDF export
O6	Investor analytics	Transaction metrics + cohort/SIP outputs
O7	Advanced analytics	VaR/CVaR, rolling Sharpe, HHI, recommender
O8	Documentation and reproducibility	Report + README + Git repository

3.1 Functional Requirements

The platform must retain raw data separately from processed data, validate key identifiers, produce cleaned datasets and load the results into SQLite. It must support queries by fund, fund house, category and time. The dashboard must allow interactive filtering and provide a drill-through path to scheme-level details.

3.2 Data Requirements

- Consistent AMFI-code identifiers across fund and NAV datasets.
- Valid dates for daily and monthly observations.
- Positive NAV and transaction amounts.
- Standardized transaction types.
- Numerically usable performance and risk fields.
- Clearly identified benchmark mappings.
- Explicit classification of synthetic/project-generated datasets.

3.3 Reproducibility Requirements

Scripts are organized by pipeline stage and analytical purpose. Processed outputs are stored separately, SQL objects are maintained under sql/, notebooks under notebooks/, and final presentation assets under dashboard/ and reports/. This structure supports re-running individual stages without mixing source files and outputs.

4. Data Sources and Dataset Description

The project uses public Indian mutual fund information from AMFI India and mfapi.in together with benchmark/public market data and constructed datasets needed for investor and portfolio analysis.

Dataset	Records	Purpose
01_fund_master.csv	40	Scheme metadata and classification
02_nav_history.csv	46,000	Historical NAV time series
03_aum_by_fund_house.csv	90	Fund-house AUM and scheme counts
04_monthly_sip_inflows.csv	48	Monthly SIP indicators
05_category_inflows.csv	144	Category-level net inflows
06_industry_folio_count.csv	21	Folio count and composition
07_scheme_performance.csv	40	Performance/risk metrics
08_investor_transactions.csv	32,778	Investor transactions and demographics
09_portfolio_holdings.csv	322	Constructed holdings and sectors
10_benchmark_indices.csv	8,050	Benchmark/index history

4.1 Fund Master

The fund master contains AMFI code, fund house, scheme name, category, sub-category, plan, launch date, benchmark, expense ratio, exit load, minimum investment amounts, fund manager, risk category and SEBI category code. It acts as the main scheme dimension for the analytical model.

4.2 NAV History

The NAV dataset contains AMFI code, date and NAV. It is the primary input for time-series return and risk calculations. Cleaning ensured dates were parsed and observations were ordered within each fund.

4.3 Investor Transactions

The transaction dataset contains investor ID, transaction date, AMFI code, transaction type, amount, location, city tier, age group, gender, annual income, payment mode and KYC status. These records are synthetic/project-generated and are used to demonstrate investor analytics.

5. Data Authenticity and ETL Pipeline

5.1 Data Provenance

Data Category	Treatment in Project
Fund metadata and AMFI codes	Public / real-source based
Historical NAV	Public-source based
Benchmark data	Public benchmark data
AUM anchors	Public-source based
Investor transactions	Synthetic / project-generated
Portfolio holdings	Constructed / project-generated
Sector weights	Derived from constructed holdings
Category inflows	Constructed for capstone where applicable

5.2 ETL Flow

RAW FILES → INGESTION → VALIDATION → CLEANING → PROCESSED CSV → SQLITE → ANALYTICS → POWER BI

5.3 Ingestion

The ingestion stage organized ten raw datasets and tested live NAV retrieval for selected schemes. Six selected schemes had live-fetch files prepared. Some API calls experienced connection timeouts; therefore, the stable prepared NAV dataset was retained as the primary analytical source.

5.4 Validation

AMFI code validation returned 40 fund-master codes, 40 NAV codes and zero missing fund-master codes. This check was important because AMFI code is the principal link between scheme metadata and time-series observations.

5.5 Cleaning

- NAV dates converted to valid date values and observations sorted by fund and date.
- Duplicate fund-date combinations checked with zero duplicates.
- NAV values checked for non-positive observations.
- Investor transaction types standardized to SIP, Lumpsum and Redemption.
- Transaction amounts and dates validated.
- KYC values standardized.
- Scheme performance numeric fields converted and checked.

6. ETL Implementation Details

6.1 NAV Cleaning Results

The NAV cleaning process started with 46,000 observations. Date conversion and sorting were applied by AMFI code. Duplicate fund-date checks returned zero duplicates, and no non-positive NAV observations remained after validation. The cleaned output was saved as 02_nav_history_clean.csv.

6.2 Transaction Cleaning Results

Transaction Type	Count
SIP	19,716
Lumpsum	8,095
Redemption	4,967
Total	32,778

KYC validation resulted in 30,146 Verified records and 2,632 Pending records. The final transaction dataset retained 13 columns and 32,778 rows.

6.3 Performance Cleaning

The scheme-performance dataset contained 40 records and 19 columns. Numeric fields were converted for calculation. Expense ratios were within the checked range of 0.1% to 2.5%, with zero observations outside that range.

6.4 Processed Layer

The processed directory provides a clean boundary between raw source files and downstream analytics. This is useful for reproducibility because dashboard and notebook workflows can use validated files without altering the original source datasets.

6.5 Error Handling and Practical Considerations

The development process included package/environment validation, API timeout handling and database verification. Live retrieval was treated as a supplementary ingestion feature rather than a dependency for every analytical run. This makes the project more robust when external APIs are temporarily unavailable.

7. Database Design and Star Schema

The SQLite database provides the persistent relational layer of the platform. The design uses dimensions for descriptive attributes and fact tables for measurable events or observations.

Table	Type	Rows	Role
dim_fund	Dimension	40	Scheme master
dim_date	Dimension	1,297	Calendar dimension
fact_nav	Fact	46,000	Daily NAV
fact_transactions	Fact	32,778	Investor transactions
fact_performance	Fact	40	Scheme metrics
fact_aum	Fact	90	Fund-house AUM

7.1 Dimension Design

dim_fund stores one record per scheme and contains the AMFI code used as the business identifier. dim_date provides calendar attributes for time filtering and consistent SQL analysis.

7.2 Fact Design

fact_nav stores daily market observations; fact_transactions stores investor events; fact_performance stores calculated or prepared scheme metrics; and fact_aum stores fund-house-level AUM observations.

7.3 Integrity Checks

Database verification confirmed the expected tables and key relationships. The final database was created at db/bluestock_mf.db and contains the row counts shown above.

7.4 Analytical SQL

Ten SQL queries were implemented to demonstrate business analysis. Examples include identifying funds by AUM, filtering schemes by expense ratio, comparing three-year performance by category and examining funds with stronger Sharpe ratios.

8. Exploratory Data Analysis — Industry Trends

8.1 AUM Trend

Year-end industry AUM values in the project dataset were ₹30.90 lakh crore in 2022, ₹37.42 lakh crore in 2023, ₹52.68 lakh crore in 2024 and ₹62.74 lakh crore in 2025. These values are industry/fund-house-level observations and should not be confused with the AUM of only the 40 selected schemes.

8.2 SIP Inflow

The latest monthly SIP inflow in the prepared industry dataset is ₹31,002 crore for December 2025. This indicator is presented prominently in the Power BI dashboard.

8.3 Folio Growth

Total folios increased from 13.26 crore to 26.12 crore across the available series. The calculated growth is 96.98%. Folio growth provides a complementary participation indicator alongside SIP flows.

8.4 Scheme and Fund-House Context

The fund master covers 40 schemes across 10 fund houses. The industry AUM dataset contains a broader fund-house view, and the dashboard uses this dataset to show top fund houses by AUM.

8.5 Interpretation

The trend analysis is descriptive rather than predictive. Increasing AUM or SIP activity can reflect a combination of market movement, net flows and changing participation. The project does not attribute causality to any single factor without additional evidence.

9. Exploratory Data Analysis — Investors and Diversification

9.1 Age Distribution

Age Group	Share
26–35	41.07%
36–45	24.85%
18–25	15.00%
46–55	11.53%
56+	7.55%

The 26–35 group is the largest age segment in the synthetic transaction dataset. This observation describes the constructed dataset and should not be interpreted as a representative estimate of the Indian mutual fund investor population.

9.2 Geography

The top states by SIP amount in the transaction analysis are Madhya Pradesh, Punjab, Telangana, Tamil Nadu and Gujarat. T30 accounts for 66.72% and B30 for 33.28% of the analyzed transaction dataset.

9.3 Correlation Analysis

Ten selected funds were compared using 45 valid unique pairwise correlations. The highest observed correlation was 0.0456 between HDFC Short Term Debt and HDFC Top 100. The lowest was -0.0935 between HDFC Top 100 and UTI Nifty 50. The average pairwise correlation was approximately -0.004.

9.4 Why Correlation Was Included

Correlation provides a diversification-oriented perspective by showing how similarly selected return series move relative to each other. The analysis does not establish that a portfolio is diversified solely because pairwise correlations are low; position sizes, sector exposure and other risk characteristics also matter.

10. Performance and Risk Analysis

The performance notebook converts historical NAV data into return and risk indicators. Since the available NAV period runs from January 2022 to May 2026, the project does not claim a genuine five-year CAGR. Instead, it reports CAGR-till-date.

10.1 Risk Metrics

Metric	Purpose
CAGR-till-date	Annualized historical growth over available period
Sharpe Ratio	Risk-adjusted return using assumed 6% risk-free rate
Sortino Ratio	Risk-adjusted return emphasizing downside deviation
Alpha	Benchmark-relative excess performance
Beta	Sensitivity relative to mapped benchmark
Maximum Drawdown	Largest peak-to-trough decline

10.2 Alpha/Beta Results

Alpha/beta analysis was available for 37 funds. Mean alpha was 10.94%, and 31 of 37 funds had positive alpha under the project's benchmark mapping and risk-free assumptions. Three funds lacked benchmark data: HDFC Short Term Debt, Axis Midcap and Mirae Emerging Bluechip.

10.3 Drawdown Results

Statistic	Result
Mean max drawdown	-17.87%
Median	-16.31%
Worst	SBI Small Cap Direct: -52.57%
Second-worst	Axis Small Cap Regular: -51.68%

Drawdown is included because return alone does not show the magnitude of historical losses experienced during the observed period.

11. Fund Scorecard and Comparative Framework

A multi-metric scorecard was created to provide a consistent analytical framework for comparing schemes. It combines normalized components rather than relying on a single return statistic.

Component	Weight
CAGR 1Y	20%
CAGR 3Y	20%
Sharpe	20%
Sortino	15%
Alpha	15%
Drawdown	10%

11.1 Scorecard Outputs

Fund	Score
SBI Small Cap	78.10
DSP Small Cap	74.90
Mirae Asset Large Cap	73.33
SBI Bluechip	73.15
HDFC Mid-Cap Opportunities	72.96
Mirae Asset Tax Saver	72.64
ICICI Pru Midcap	72.46
Kotak Flexicap	71.25
DSP Midcap	66.37
ABSL Frontline Equity	66.06

11.2 Handling Missing Data

Three funds have missing benchmark-relative alpha/beta analysis. Missing alpha values were assigned a neutral score of 50 within the scorecard, while other missing components were also handled neutrally. This prevents a missing field from automatically eliminating a fund from comparison.

11.3 Limitations of the Score

The score is a project-specific analytical construct. Its weights are design choices and do not represent an industry-standard fund rating. Historical metrics also do not guarantee future results.

12. Advanced Analytics

12.1 Historical VaR and CVaR

Daily returns were calculated fund by fund. VaR at the 95% level was estimated as the fifth percentile of daily returns. CVaR was calculated as the mean of returns at or below the VaR threshold. Under the project's negative-return convention, a more-negative result represents greater historical downside exposure.

12.2 Rolling 90-Day Sharpe

A rolling 90-day window was applied to five funds selected from the scorecard: SBI Small Cap, DSP Small Cap, Mirae Asset Large Cap, SBI Bluechip and HDFC Mid-Cap Opportunities. Rolling mean and standard deviation were annualized using $\sqrt{252}$.

12.3 Investor Cohorts

Cohorts were formed using each investor's first transaction year. The resulting dataset includes investor count, total investment, total redemption, net investment, average SIP amount, SIP amount, lumpsum amount, investment preference and SIP share.

12.4 SIP Continuity

For every investor with multiple SIP observations, the gap between consecutive SIP transactions was calculated. A gap greater than 35 days is classified as 'At Risk' according to the project-defined rule. The synthetic transaction window covers 2024–2025.

13. Recommendation Prototype and Sector Concentration

13.1 Rule-Based Recommendation

The recommender is a transparent rule-based prototype rather than a machine-learning model or personalized financial-advice system. It combines overall fund score (35%), risk match (25%), investment horizon (20%) and historical return (20%).

Fund	Prototype Score
Mirae Asset Large Cap Fund - Regular - Growth	93.13
SBI Bluechip Fund - Regular Plan - Growth	89.76
Kotak Flexicap Fund - Regular - Growth	85.67
ABSL Frontline Equity Fund - Regular - Growth	84.33
Nippon India Large Cap Fund - Regular - Growth	83.02

13.2 Sector Concentration

The constructed holdings dataset was analyzed by sector. The largest sector weights were Banking 19.18%, IT 13.40%, Pharma 11.98%, Automobile 9.52% and Utilities 7.81%. The top five sectors together account for 61.89%.

13.3 HHI Method

HHI is calculated by converting sector weights to proportions, squaring each proportion and summing the results. The project reports HHI on 0–1 and 0–10,000 scales and also calculates effective number of sectors as the reciprocal of HHI.

13.4 Interpretation

Sector concentration helps identify whether constructed portfolio exposure is distributed across many sectors or dominated by fewer sectors. Because the holdings are constructed for the capstone, the result should not be interpreted as an official disclosure.

14. Power BI Dashboard

The Power BI dashboard is the primary business-facing interface of the project. It combines prepared data, DAX measures and interactive filtering into a multi-page analytical experience.

14.1 Industry Overview

- Total selected-scheme AUM card.
- Latest SIP inflow card showing ₹31,002 crore.
- Latest folio card showing 26.12 crore.
- Latest number of schemes indicator.
- Industry AUM trend.
- Top ten fund houses by AUM.
- Year slicer.

14.2 Fund Performance

- Average 1-year return.
- Average 3-year return.
- Average Sharpe ratio.
- Average maximum drawdown.
- Top ten funds by scorecard.
- Top ten funds by one-year return.
- Risk versus three-year return scatter.
- Fund and fund-house slicers.

14.3 Investor Analytics

- Distinct investors and transaction-value KPIs.
- Investment versus redemption.
- Investment by age group.
- Top states by investment.
- SIP versus lumpsum by age.
- Gender breakdown and demographic filters.

14.4 SIP & Market Trends

- Latest active SIP accounts.
- Latest new SIP accounts.
- Latest SIP AUM.
- Monthly SIP inflow trend.
- Category inflow trend.
- Folio growth and composition.
- SIP YoY growth.

15. Power BI Data Model, DAX and UX

15.1 Data Model

The model uses 01_fund_master as the central fund dimension. Relationships connect AMFI codes to NAV, performance and transaction tables. DimDate connects to daily NAV, transaction and AUM dates. A separate DimMonth supports monthly SIP, category-inflow and folio datasets.

15.2 Key DAX Measures

Measure	Purpose
Total AUM (Lakh Cr)	Sum scheme AUM divided by 100,000
Latest SIP Inflow (Cr)	Latest monthly SIP inflow
Latest Folios (Cr)	Latest total folios
Latest Number of Schemes	Latest fund-house scheme count
Average 1Y Return	Average scheme 1-year return
Average 3Y Return	Average scheme 3-year return
Average Sharpe	Average scheme Sharpe
Average Max Drawdown	Average scheme drawdown
Total Investors	Distinct investor IDs
Total Investment (Cr)	SIP + lumpsum amounts
Total Redemptions (Cr)	Redemption amounts

15.3 Slicers and Navigation

Slicers were added according to page purpose. Page 2 includes fund and fund-house filters; Page 3 includes age group, city tier and fund house; Page 4 includes year filtering through DimMonth; Page 1 includes year filtering through DimDate.

15.4 Drill-Through

A Fund Details page was configured with scheme name as the drill-through field. The user can select a scheme from the performance page, open its focused details and return using the back button.

15.5 Dashboard QA

Final QA included checking card titles, chart labels, slicer behavior, page navigation, relationship behavior and visual overlap. Screenshots and a PDF export were retained as project deliverables.



MUTUAL FUND INDUSTRY OVERVIEW



SIP Inflows

Latest Number of Schemes

2K

Folios

Latest Folios (Cr)

26.12

Total AUM

Total AUM (Lakh Cr)

10.44

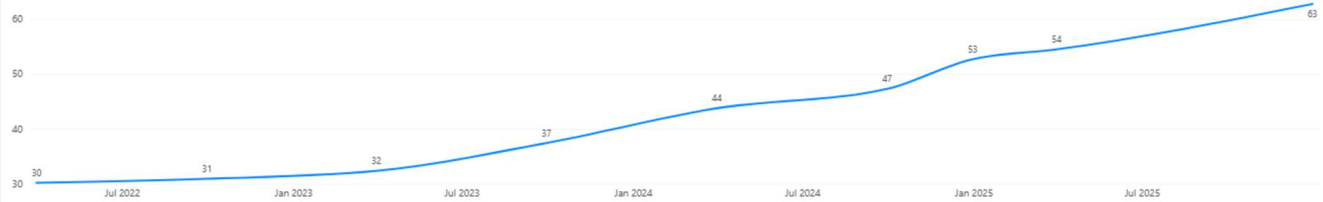
Number of Schemes

Latest SIP Inflow (Cr)

31.002K

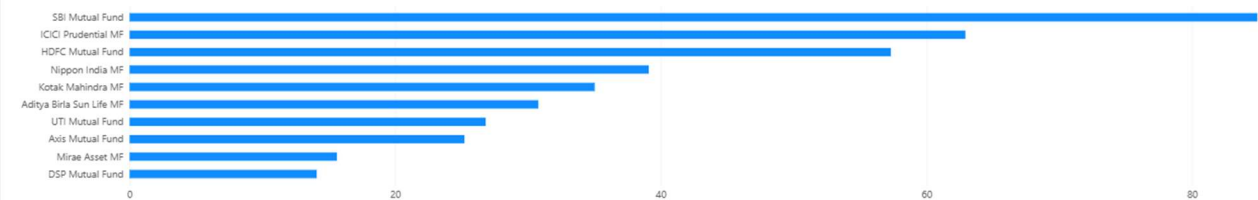
Industry AUM — Jan 2022 to Dec 2025

Sum of aum_lakh_crore by date



Top 10 Fund Houses by AUM

Sum of aum_lakh_crore by fund_house



Fund Performance

Select Fund

scheme_name

☐ ABSL Frontline Equity Fund - Regular - Growth

☐ ABSL Liquid Fund - Regular - Growth

Average 1Y Return

Average 1Y Return (%)

14.38

Average 3Y Return

Average 3Y Return (%)

14.09

Average Max Drawdown

Average Max Drawdown (%)

-19.20

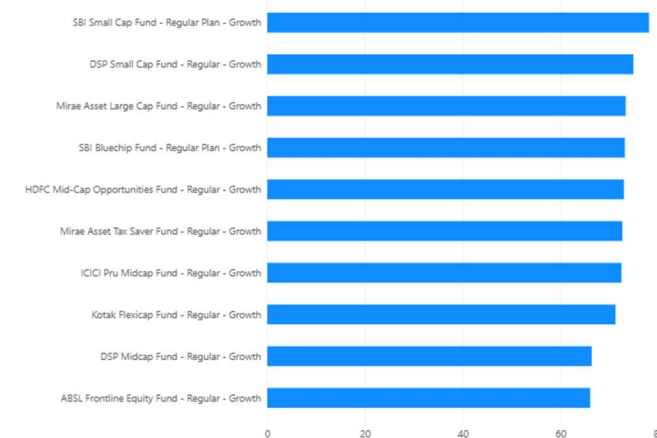
Average Sharpe Ratio

Average Sharpe Ratio

1.36

Top 10 Funds by Overall Score

Sum of Overall_Score by scheme_name

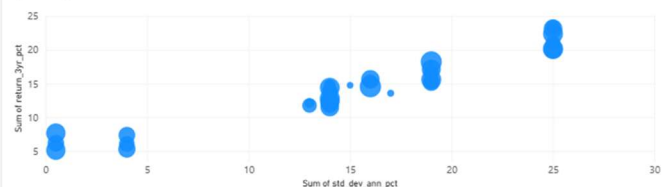


Top 10 Funds by 1-Year Return

Sum of return_1yr_pct by scheme_name



Sum of std_dev_ann_pct. Sum of return_3yr_pct and Sum of aum_crore by scheme_name





INVESTOR ANALYTICS

Total Investors

Total Investors
5K

Total Investment

Total Investment (Cr)
227.71

Total Redemptions

Total Redemptions (Cr)
124.45

Average Transaction

Average Transaction (₹)
107.44K

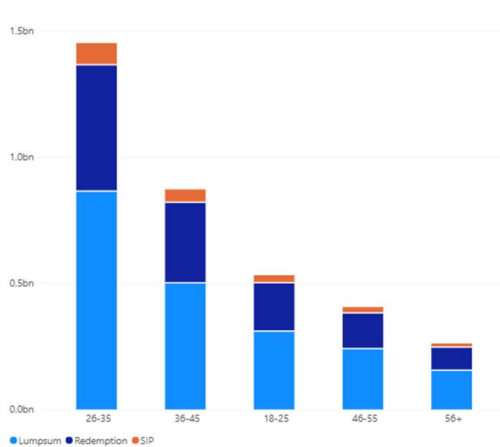
Investment vs Redemption

Sum of amount_inr by transaction_type



SIP vs Lumpsum by Age Group

Sum of amount_inr by age_group and transaction_type



City Tier

city_tier

830

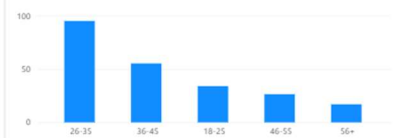
Age Group

age_group

18-25

Investment by Age Group

Total Investment (Cr) by age_group



Investment by State

Total Investment (Cr) by state



SIP & MARKET TRENDS

Year

2022

2025

SIP Inflows

Latest Number of Schemes
2K

Active SIP Accounts

Latest Active SIP Accounts (Cr)
9.35

New SIP Accounts

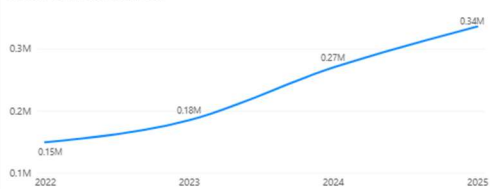
Latest New SIP Accounts (Lakh)
9.80

SIP AUM

Latest SIP AUM (Lakh Cr)
15.90

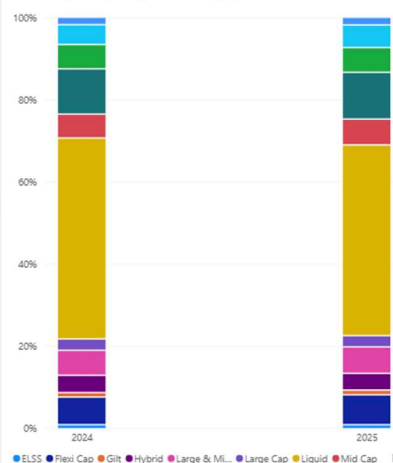
Monthly SIP Inflow Trend

Sum of sip_inflow_crore by Year



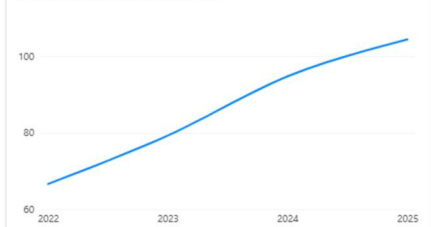
Category-wise Net Inflows

Sum of net_inflow_crore by Year and category



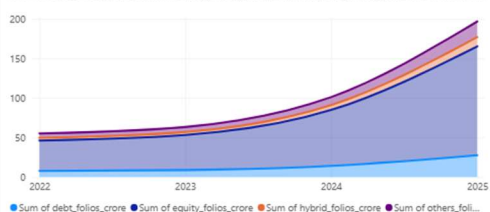
Active SIP Accounts Trend

Sum of active_sip_accounts_crore by Year

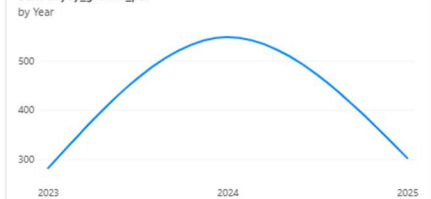


Folio Composition by Category

Sum of debt_folios_crore, Sum of equity_folios_crore, Sum of hybrid_folios_crore and Sum of others...



Sum of yoy_growth_pct by Year



16. Data Quality, Assumptions and Limitations

16.1 Quality Checks

Check	Result
Fund-master AMFI codes	40
NAV AMFI codes	40
Missing fund-master codes in NAV	0
Duplicate fund-date NAV records	0
Non-positive NAV observations	0
Clean investor transaction rows	32,778
Clean performance rows	40

16.2 Analytical Assumptions

- Annual risk-free rate assumed at 6%.
- 252 trading days used for annualization.
- 35-day gap used for SIP continuity risk classification.
- Neutral score of 50 used for selected missing score components.
- Benchmark mapping used only where compatible benchmark data were available.

16.3 Important Limitations

- Available NAV history is shorter than five complete years.
- Three funds lack benchmark data for alpha/beta.
- Investor transactions are synthetic.
- Portfolio holdings and sector weights are constructed.
- Category inflows are constructed for the capstone where applicable.
- External API calls may experience timeouts.
- Historical analytics should not be interpreted as future-performance guarantees.

16.4 Responsible Interpretation

The platform is an analytics demonstration. The recommendation prototype and scorecard should be treated as transparent project logic rather than personalized financial advice. Any production financial application would require stronger data governance, suitability controls, regulatory review and validated real-world datasets.

17. Appendix

17.1 Final Project Folder Structure

```
bluestock_mf_capstone/  
├── data/  
│   ├── raw/  
│   └── processed/  
├── db/  
│   └── bluestock_mf.db  
├── dashboard/  
│   └── Mutual_Fund_Analytics_Dashboard.pbix  
├── notebooks/  
│   ├── 03_eda_analysis.ipynb  
│   ├── 04_performance.ipynb  
│   └── 05_Advanced_Analytics.ipynb  
├── reports/  
│   ├── Final_Report.md  
│   ├── Mutual_Fund_Analytics_Dashboard.pdf  
│   ├── data_dictionary.md  
│   ├── industry_overview.png  
│   ├── fund_performance.png  
│   ├── investor_analytics.png  
│   └── sip_market_trends.png  
├── scripts/  
│   ├── data_ingestion.py  
│   ├── live_nav_fetch.py  
│   ├── fund_master_analysis.py  
│   ├── validate_amfi_codes.py  
│   ├── clean_nav_history.py  
│   ├── clean_investor_transactions.py  
│   ├── clean_scheme_performance.py  
│   ├── load_database.py  
│   ├── verify_database.py  
│   ├── run_all_queries.py  
│   ├── recommender.py  
│   └── sector_hhi.py  
├── sql/  
│   ├── schema.sql  
│   └── queries.sql  
├── requirements.txt  
├── README.md  
└── .gitignore
```

17.2 Key Dataset Counts

Dataset	Records
Fund Master	40

NAV History	46,000
Investor Transactions	32,778
Scheme Performance	40
AUM by Fund House	90
Fund Holdings	322
Benchmark Data	8,050

17.3 Database Row Counts

Table	Rows
dim_fund	40
dim_date	1,297
fact_nav	46,000
fact_transactions	32,778
fact_performance	40
fact_aum	90

17.4 Selected Analytical Metrics

Metric	Result
Fund schemes analyzed	40
Fund houses	10
Latest SIP inflow	₹31,002 Cr
Latest folios	26.12 Cr
2025 fund-house AUM	₹62.74 Lakh Cr
Selected-scheme AUM	Approximately ₹8.81 Lakh Cr
Folio growth	96.98%
Top 5 sector concentration	61.89%
Transaction records	32,778
SIP transactions	19,716
Lumpsum transactions	8,095
Redemption transactions	4,967

18 Final Conclusion

The completed project demonstrates an end-to-end mutual fund analytics workflow that integrates data engineering, database management, financial analytics and business intelligence. The platform is structured so

that the raw data layer, cleaned data layer, database, analytical notebooks and dashboard remain logically separated.

The final outputs provide a practical demonstration of how mutual fund information can be transformed into decision-support dashboards while clearly documenting data limitations, assumptions and synthetic components. The architecture can be extended toward automated refresh, longer historical data, stronger benchmark attribution, production-grade data governance and advanced investor analytics.

End of Final Project Report